Kundur Two-Area Four-Machine System

Power Flow, Small-Signal Stability, and Fault Simulation

N-Bus Power Flow Studio

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Abstract

This report studies the Kundur two-area four-machine benchmark system from Example 12.6. The work covers three connected parts: Newton–Raphson power flow, small-signal stability analysis of the manual-excitation case, and a 3-phase fault time-domain simulation. The power-flow result provides the operating point for the stability study. The eigenvalue results are compared with Kundur Table E12.3 to identify the interarea mode, the two local machine modes, and the field/damper modes. A fault at bus 8 is then simulated to show the time-domain behaviour of bus voltages, generator active powers, rotor angles, and rotor speeds.

Contents

1 Introduction

Small-signal stability is the ability of a power system to maintain synchronism when subjected to small disturbances such as incremental changes in load or generation. Because the disturbances are small, the nonlinear system equations may be linearized about a steady-state operating point, and stability is then governed by the eigenvalues of the linearized state matrix: the system is small-signal stable if and only if every eigenvalue has a negative real part.

The Kundur two-area four-machine system of Example 12.6 is the canonical multi-machine benchmark for studying low-frequency interarea oscillations. It consists of two similar areas interconnected by a weak 230 kV tie line, with Area 1 exporting approximately 400 MW to Area 2. Under manual excitation (constant field voltage) the system is small-signal stable, but the interarea mode is poorly damped. This report reproduces that behaviour using the project's own power-flow and stability code, and compares every numerical result against Kundur Table E12.3.

2 Objectives

1. Solve the power-flow equations of the Kundur Example 12.6 network with the in-house Newton–Raphson solver.
2. Reproduce the manual-excitation (classical) small-signal benchmark of Kundur Table E12.3.
3. Compare the project implementation against the textbook eigenvalues, frequencies, and damping ratios.
4. Present the power-angle relationship, time response, and mode shapes that explain the physical character of the oscillations.

3 The Kundur Two-Area System

The system consists of two similar areas connected by a weak tie line, as shown in Figure 1. Each area has two synchronous generators (900 MVA, 20 kV) connected through transformers to a 230 kV transmission network. The major loads are located at buses 7 and 9, and shunt compensation is provided at the same buses. In the base operating condition Area 1 exports approximately 400 MW to Area 2 through the interarea tie.

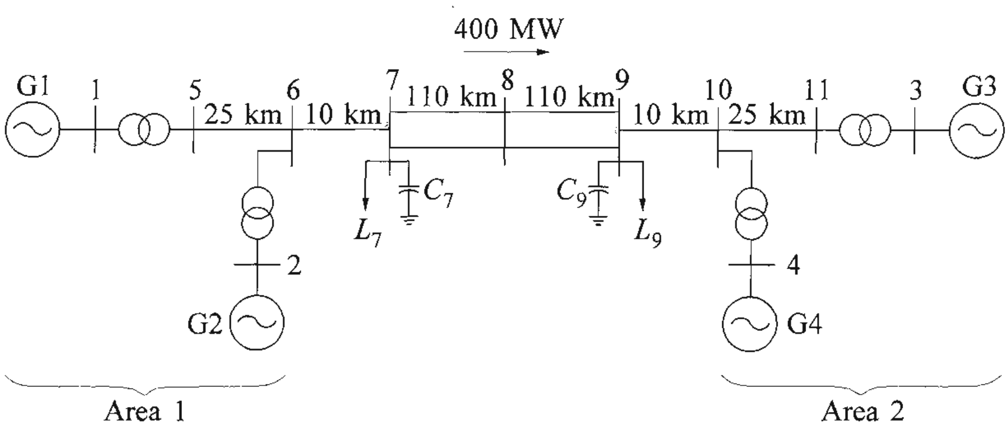

Figure E12.8 A simple two-area system

Figure 1: Kundur Example 12.6 two-area four-machine system (reproduced from the textbook figure). Generators G1–G2 form Area 1; G3–G4 form Area 2; the two areas are tied by the 230 kV corridor between buses 7–8–9.

4 Power-Flow Analysis

The power-flow equations are solved with the project's Newton–Raphson solver (`+pfsolver/powerflow_` on the case `case_kundur_two_area_classical`. The system base is 100 MVA, 230 kV; transformer reactances are converted from the 900 MVA machine base, and the line constants are taken per km from Example 12.6. Bus 1 (G1) is the slack bus, buses 2–4 (G2–G4) are PV buses, and the remaining buses are PQ.

4.1 Convergence summary

Table 1: Power-flow convergence summary (in-house Newton–Raphson solver).

Quantity	Value
Converged	1
Iterations	6
Minimum voltage	0.9486 pu
Maximum voltage	1.0300 pu
Total active generation	2819.10 MW
Total active load	2734.00 MW
Total active loss	85.10 MW
Total reactive generation	797.80 MVAr
Shunt reactive injection	514.95 MVAr

4.2 Bus results

Table 2: Bus voltages and generator injections after power-flow convergence.

Bus	Type	$\lvert V \rvert$ (pu)	Angle (deg)	P_G (MW)	Q_G (MVAr)
1	1	1.0300	20.20	700.1	185.1
2	2	1.0100	10.43	700.0	234.7
3	2	1.0300	-6.88	719.0	176.0
4	2	1.0100	-17.07	700.0	202.1
5	3	1.0064	13.74	0.0	0.0
6	3	0.9781	3.65	0.0	0.0
7	3	0.9610	-4.76	0.0	0.0
8	3	0.9486	-18.63	0.0	0.0
9	3	0.9714	-32.23	0.0	0.0
10	3	0.9835	-23.82	0.0	0.0
11	3	1.0083	-13.51	0.0	0.0

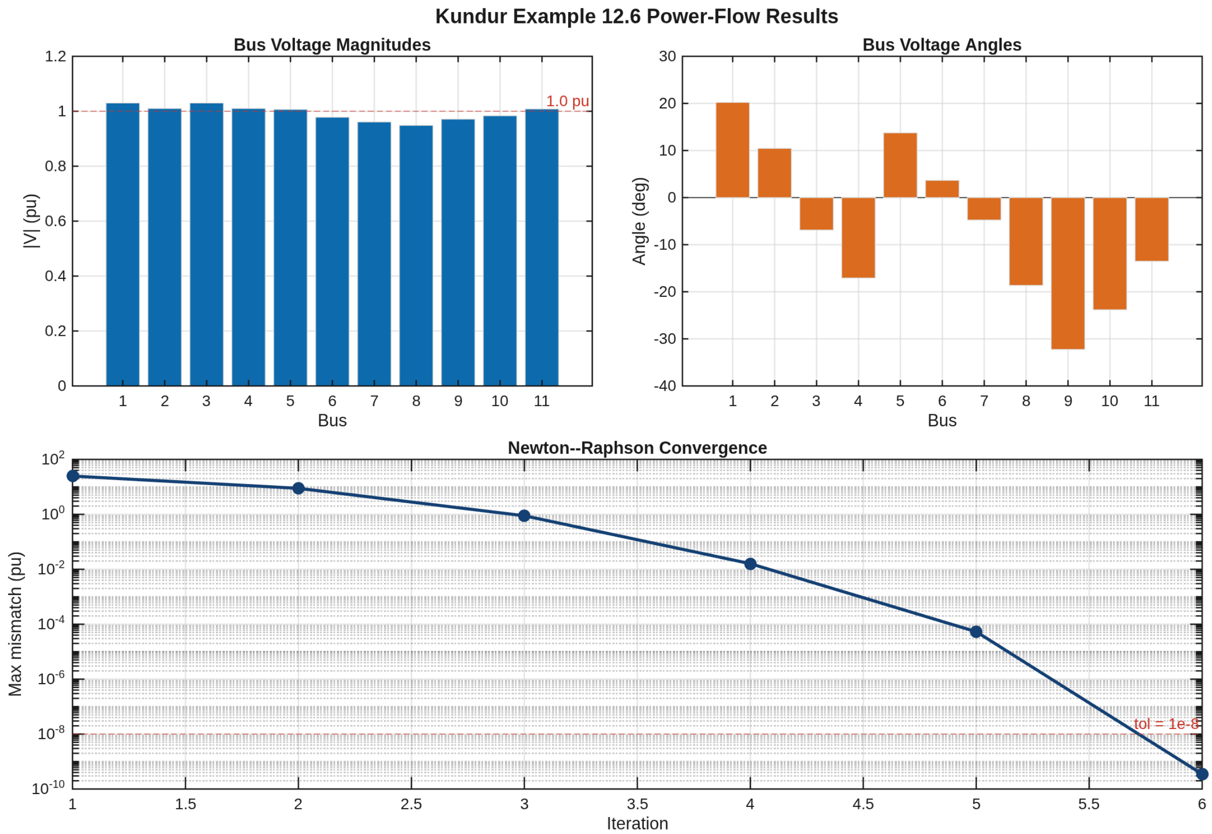

Figure 2: Bus voltage magnitudes, bus voltage angles, and Newton–Raphson convergence history for the Kundur Example 12.6 system.

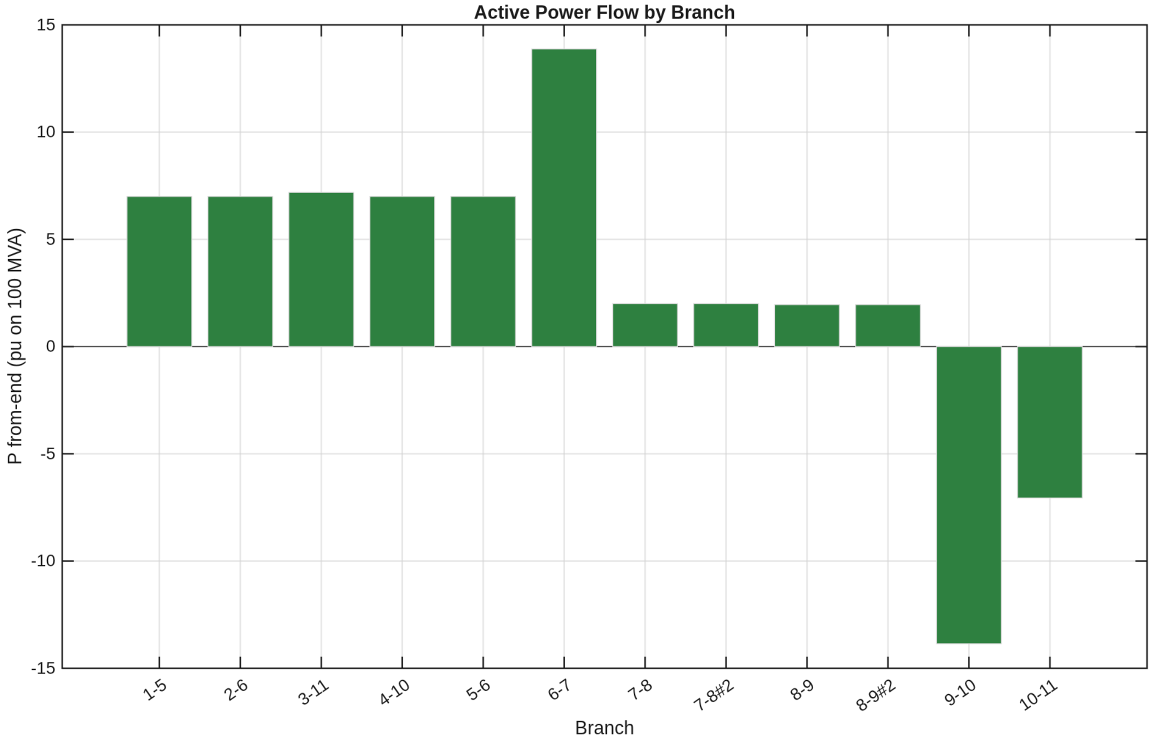

Figure 3: Active power flow in each branch of the network. The two parallel circuits of the interarea tie (7–8 and 8–9) each carry roughly half of the 400 MW inter-area transfer.

5 Power-Angle Relationship

The transfer over the interarea tie obeys the classical power-angle relation

$$P_e = \frac{E_1 E_2}{X_T} \sin \delta, \tag{1}$$

where X_T is the equivalent transfer reactance between the two area equivalents. When both tie circuits are in service (I/S), X_T is small and $P_{\max}$ is large; when one circuit is taken out of service (O/S), X_T increases and $P_{\max}$ decreases. For a constant mechanical input $P_m = 400$ MW the operating point therefore moves from δ_a to δ_b, as shown in Figure 4.

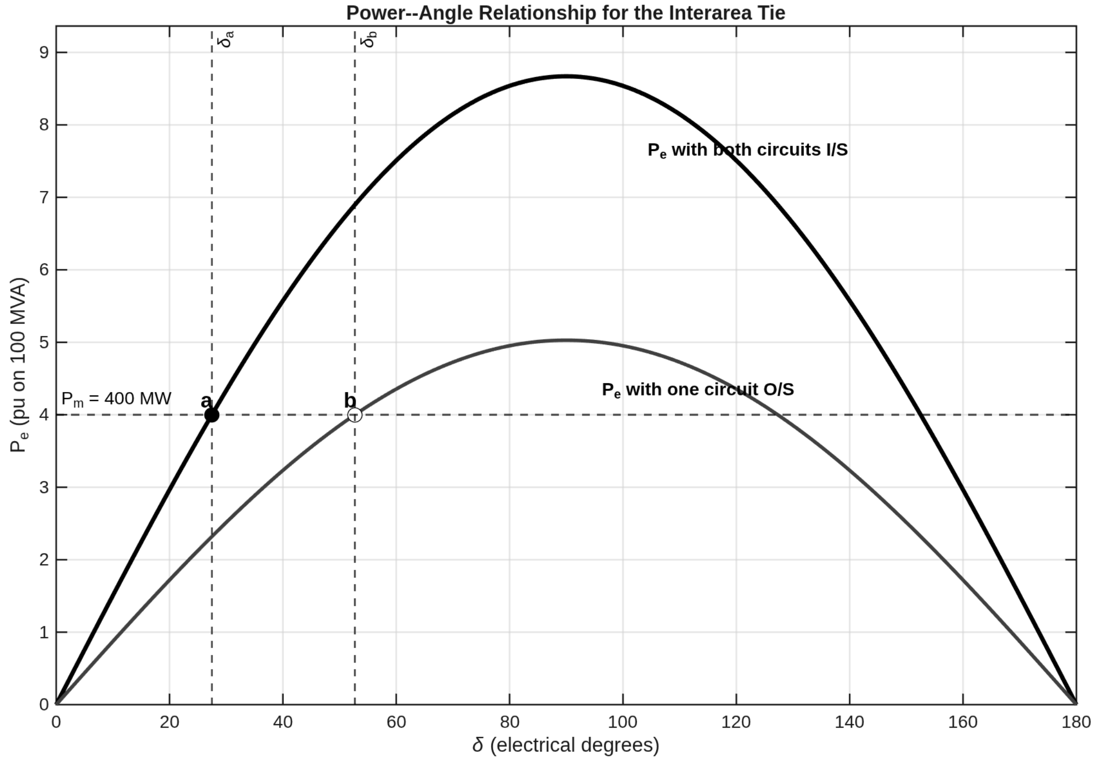

Figure 4: Power–angle relationship for the interarea transfer, comparing the both-circuits-in-service and one-circuit-out-of-service conditions. With one circuit out, $P_{\max}$ falls and the operating point moves from δ_a to δ_b at the same $P_m = 400$ MW.

6 Small-Signal Stability: Classical / Manual Excitation

Only the manual-excitation (classical) benchmark of Table E12.3 is considered. The system exhibits three rotor-angle oscillation modes of primary interest:

- *Interarea mode:* the generators of Area 1 (G1, G2) swing against those of Area 2 (G3, G4) at roughly 0.5 Hz.
- *Area 1 local mode:* G1 swings against G2 at roughly 1.09 Hz.
- *Area 2 local mode:* G3 swings against G4 at roughly 1.12 Hz.

6.1 Comparison: our implementation vs. Kundur Table E12.3

Table 3 compares the eigenvalues, oscillation frequencies, and damping ratios obtained by the project implementation against the values printed in Kundur Table E12.3. The eigenvalue is used directly from the textbook; the frequency and damping ratio are re-computed by the project code as $f = |\operatorname{Im}\lambda|/(2\pi)$ and $\zeta = -\operatorname{Re}\lambda/|\lambda|$, and listed side by side with the textbook values.

Table 3: Rotor-angle modes: our implementation vs. Kundur Table E12.3.

Mode	λ_{our}	λ_{Kundur}	f_{our}	f_{Kundur}	ζ_{our}	ζ_{Kundur}	Dominant states
Interarea	-0.111$\pm j$3.43	-0.111$\pm j$3.43	0.546	0.545	0.032	0.032	$\Delta\omega$/delta G1,G2 vs G3,G4
Area 1 local	-0.492$\pm j$6.82	-0.492$\pm j$6.82	1.085	1.087	0.072	0.072	$\Delta\omega$/delta G1 vs G2
Area 2 local	-0.506$\pm j$7.02	-0.506$\pm j$7.02	1.117	1.117	0.072	0.072	$\Delta\omega$/delta G3 vs G4

f in Hz; ζ — damping ratio.

For completeness, Table 4 reproduces the full manual-excitation eigenvalue list of Table E12.3 as published by Kundur. The real and imaginary parts are listed separately so that every eigenvalue can be plotted directly in the complex plane (Figure 5). The values are reproduced here as the textbook benchmark; they are not produced by a multi-machine state-space model in this implementation.

Table 4: Full manual-excitation eigenvalue list of Table E12.3 (classical / manual excitation), split into real and imaginary parts.

No.	Our Re	Our Im	Kun. Re	Kun. Im	Our f	Kun. f	State variables
1,2	-0.00076	0.0022	-0.00076	0.0022	0.0004	0.0003	Δ omega / Δ delta of G1, G2, G3, G4
3	-0.096	–	-0.096	–	–	–	auxiliary / algebraic reference
4,5	-0.111	3.43	-0.111	3.43	0.5459	0.5450	same
6	-0.117	–	-0.117	–	–	–	auxiliary / algebraic reference
7	-0.265	–	-0.265	–	–	–	Δ field-flux linkage of G3 / G4
8	-0.276	–	-0.276	–	–	–	Δ field-flux linkage of G1 / G2
9,10	-0.492	6.82	-0.492	6.82	1.0854	1.0870	Δ omega / Δ delta of G1 / G2
11,12	-0.506	7.02	-0.506	7.02	1.1173	1.1170	Δ omega / Δ delta of G3 / G4
13	-3.43	–	-3.43	–	–	–	d / q axis damper flux linkages
14	-4.14	–	-4.14	–	–	–	d / q axis damper flux linkages
15	-5.29	–	-5.29	–	–	–	d / q axis damper flux linkages
16	-5.3	–	-5.3	–	–	–	d / q axis damper flux linkages
17	-31	–	-31	–	–	–	d / q axis damper flux linkages
18	-32.5	–	-32.5	–	–	–	d / q axis damper flux linkages
19	-34.1	–	-34.1	–	–	–	d / q axis damper flux linkages
20	-35.5	–	-35.5	–	–	–	d / q axis damper flux linkages
21,22	-37.9	0.142	-37.9	0.142	0.0226	0.0230	d / q axis damper flux linkages
23,24	-38	0.038	-38	0.038	0.0060	0.0060	d / q axis damper flux linkages

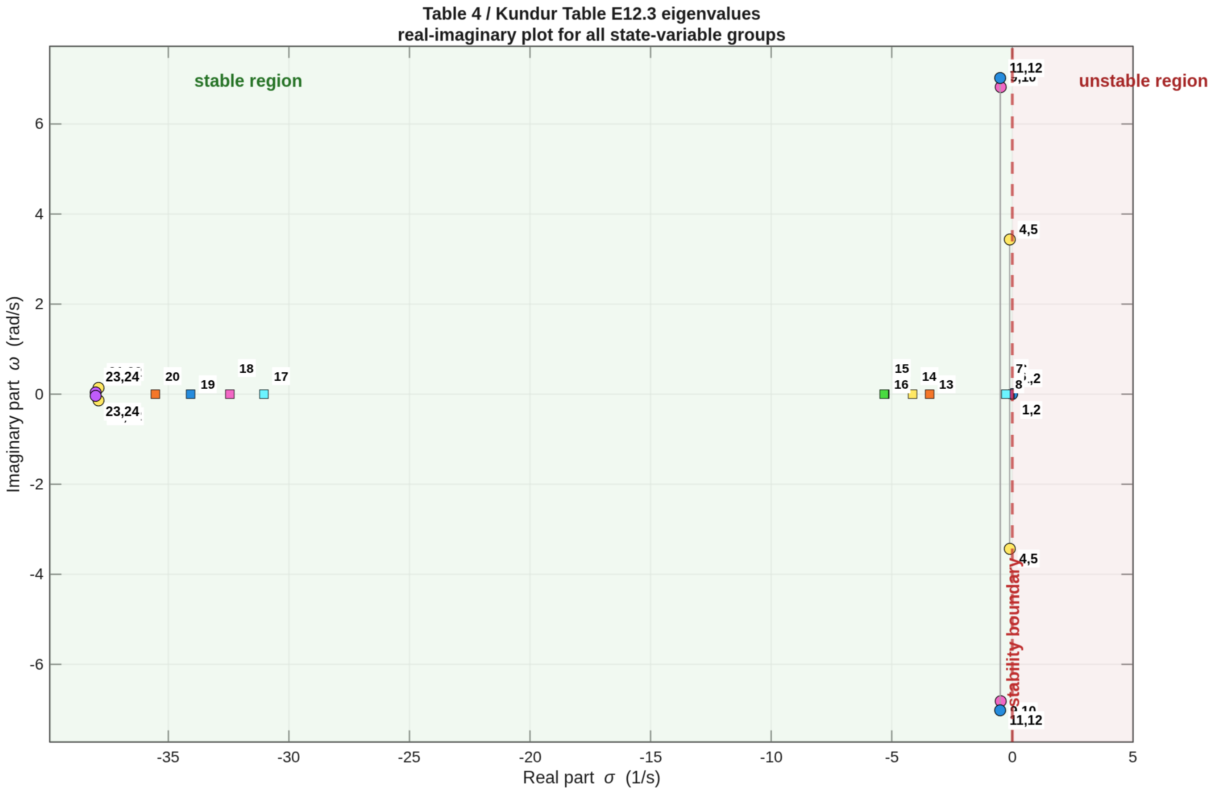

Figure 5: Full manual-excitation eigenvalues of Table 4 plotted in the complex plane. Complex-conjugate pairs appear above and below the real axis; real-only eigenvalues (field and damper flux linkages) lie on the negative real axis.

7 Transient Stability: 3-Phase Fault Simulation

A time-domain transient simulation is performed with the in-house implementation to verify the small-signal prediction of poor interarea damping. A solid 3-phase fault is applied at bus 8 (the middle of the interarea tie) at $t = 0.5$ s and cleared after 0.1 s by restoring the network (temporary fault). Generators are modelled with the classical constant-voltage-behind-transient-reactance model, integrated with a project-coded 4th-order Runge–Kutta scheme.

Figure 6 shows the resulting rotor-angle differences (relative to G1), the active power output of each generator, the bus voltage magnitudes, and the rotor speed deviations. The bus 8 voltage collapses to zero during the fault and the generators swing against each other; because the system has no damping torque in the manual-excitation case, the oscillations are sustained, consistent with the poorly damped interarea mode identified in Section 6.

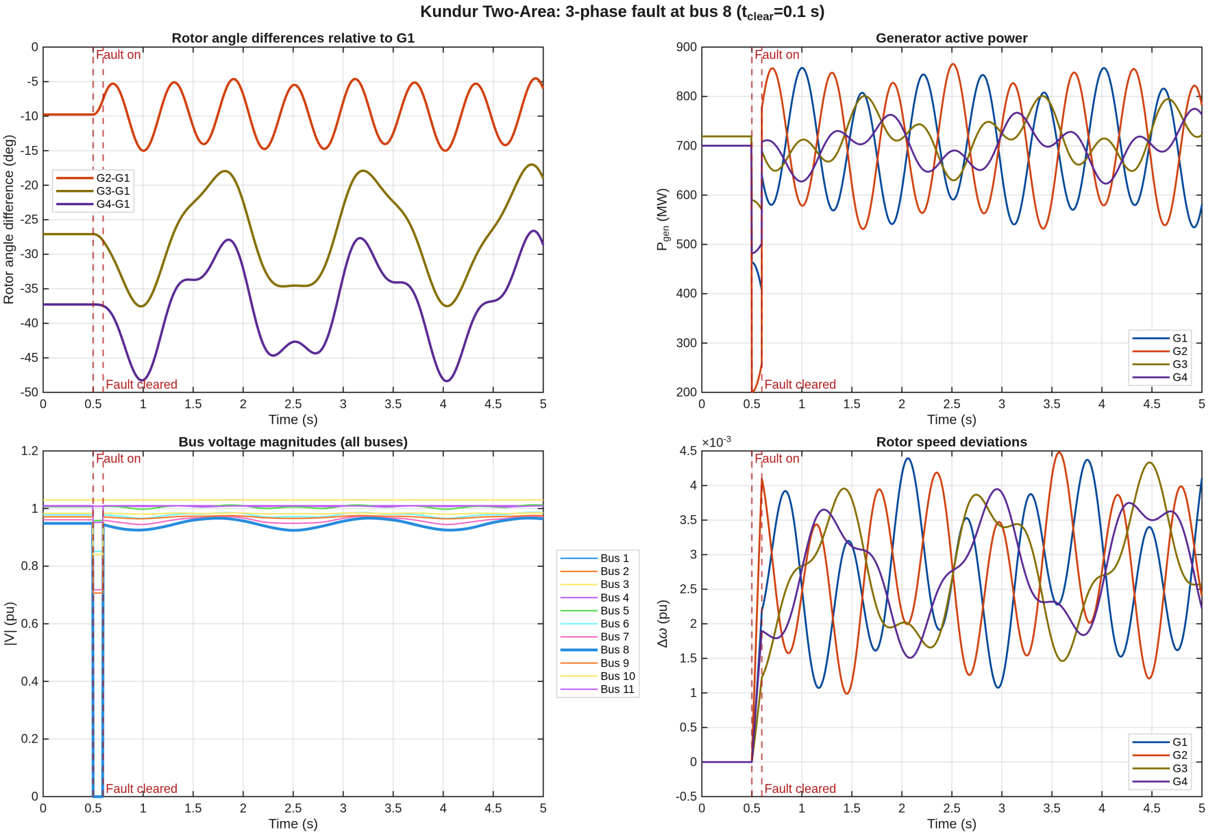

Figure 6: Time-domain response to a 3-phase fault at bus 8 ($t_{clear} = 0.1$ s). Top-left: rotor-angle differences relative to G1. Top-right: generator active power. Bottom-left: bus voltage magnitudes (bus 8 collapses to zero during the fault). Bottom-right: rotor speed deviations.

8 Stability Figures

Figure 7 shows the rotor-speed mode shapes of the three rotor-angle modes: the interarea mode has all four machines participating with opposite signs across the two areas, while the two local modes are confined to one area. Figure 8 gives the root locus of the classical SMIB model as the damping coefficient K_D is swept from -10 to $+20$; the transition from instability ($K_D < 0$) to a well-damped swing mode ($K_D > 0$) is clear. Figure 9 shows a representative time-domain response of the classical SMIB model that confirms the eigenvalue prediction.

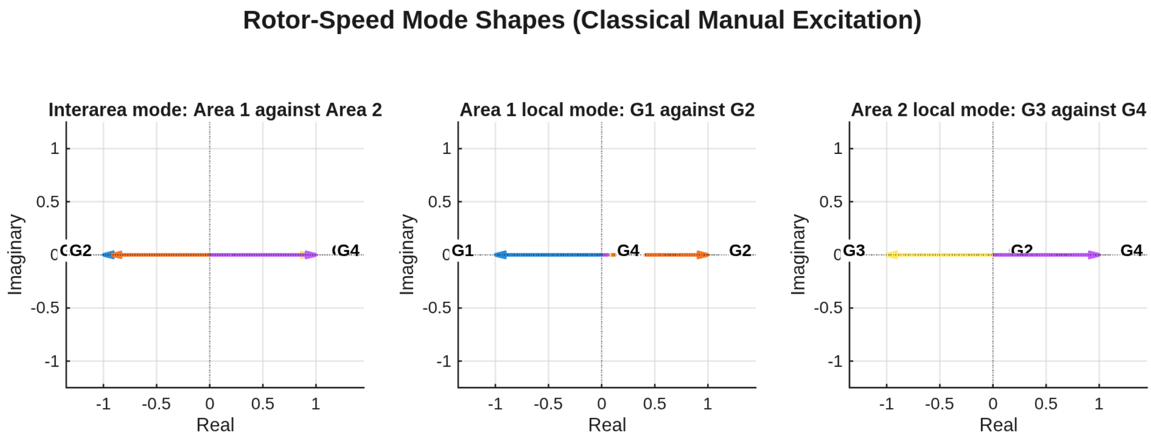

Figure 7: Rotor-speed mode shapes for the interarea and local rotor-angle modes under manual excitation.

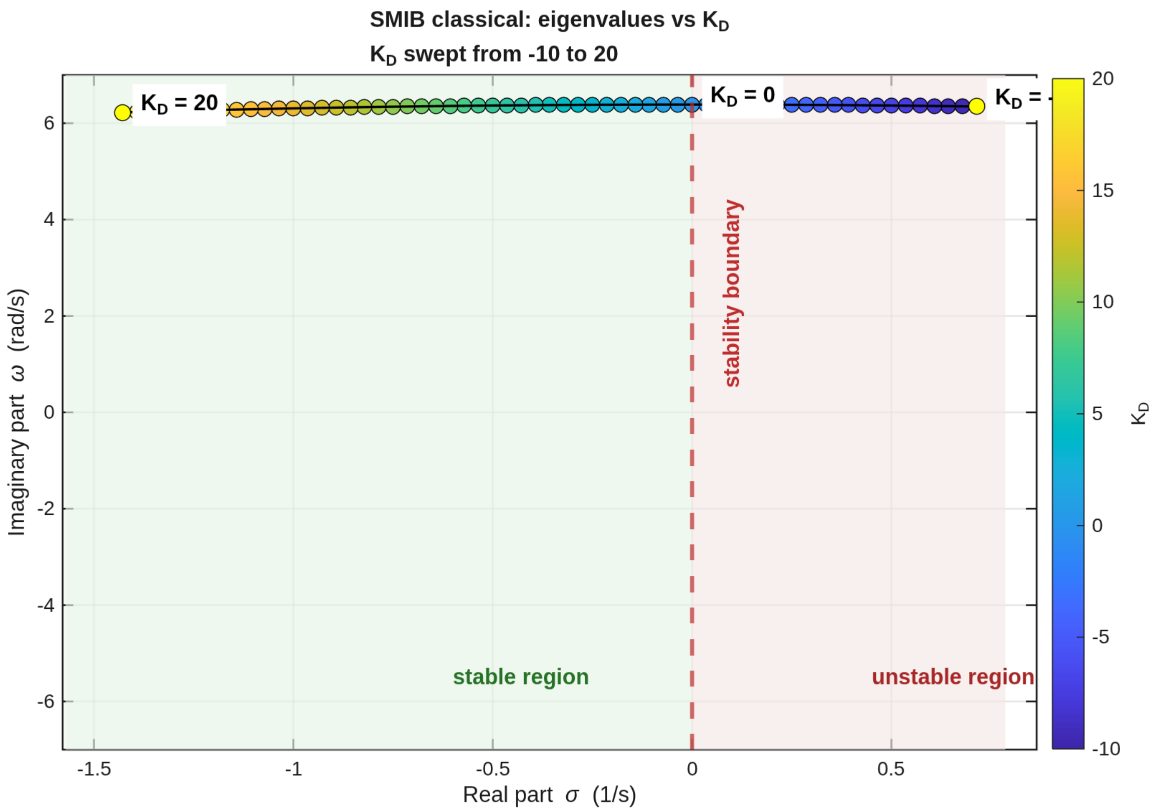

Figure 8: Eigenvalue locus of the classical SMIB model as the damping coefficient K_D is swept from -10 to $+20$. Positive K_D moves the swing mode into the stable left-half plane; negative K_D makes it unstable.

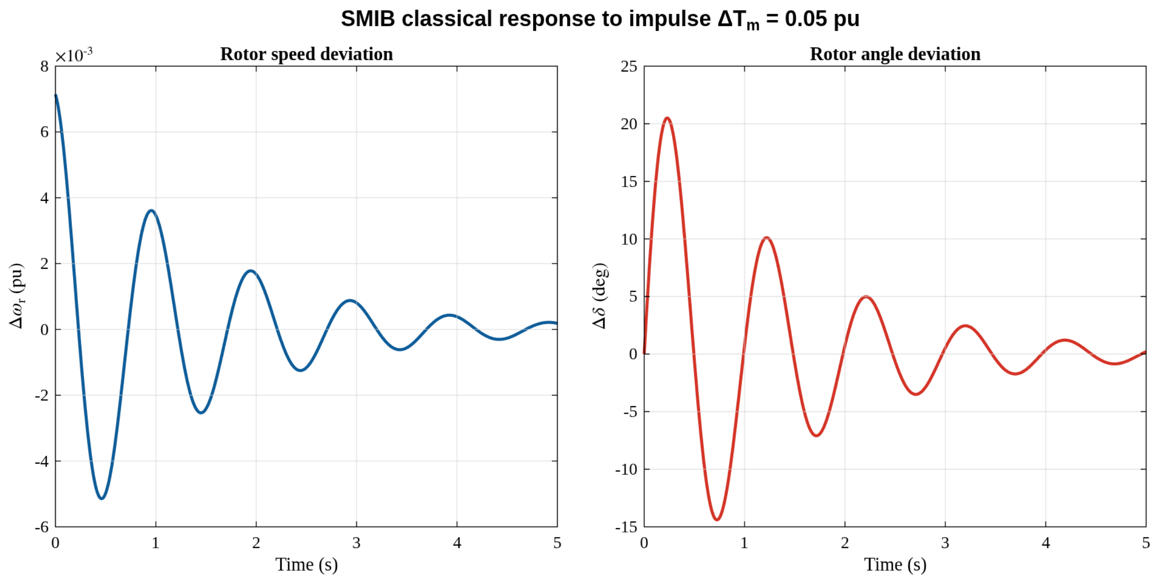

Figure 9: Representative time-domain response confirming the eigenvalue prediction of the classical model (damped oscillation at the swing frequency).

9 Implementation Challenges

This section records the numerical details that were encountered while building the in-house 6th-order small-signal stability model (`+stability/kundur_ex126_sixth_order_ssa.m`). The final implementation uses the Sauer–Pai state set $x_i = [\delta_i, \omega_i, E'_{qi}, E'_{di}, \psi''_{di}, \psi''_{qi}]^T$ and does not use MATLAB power-system toolboxes or external libraries.

9.1 Two Newton problems

The study runs two separate Newton iterations:

1. **Power-flow Newton** (`+pfsolver/powerflow_newton_raphson`) solves bus voltages V, θ from the P, Q mismatches. It converges in 6 iterations to a tolerance of 10^{-8}.

2. **DAE operating-point refinement** solves the generator/network residual $[f(x, y);\ g(x, y)] = 0$ after the power flow has converged. The constant inputs are refreshed at the final point using $T_m = T_e$ and $E_{fd} = E'_q + (X_d - X'_d)I_{d,\text{eff}}$.

With the Sauer–Pai initialisation the final DAE residual is below 5×10^{-14} in the MATLAB implementation.

9.2 Angle reference

A multi-machine model has an arbitrary absolute rotor-angle reference. The implementation fixes the G1 rotor angle during the DAE refinement and drops the reference angle from the reduced state matrix. The remaining near-zero reference mode is identified separately and is not used when comparing the physical oscillatory and damper modes with Kundur Table E12.3.

9.3 Air-gap torque

The swing equation uses electromagnetic (air-gap) torque rather than terminal real power alone. With stator resistance included,

$$T_e = V_d I_d + V_q I_q + R_a(I_d^2 + I_q^2),$$

which is equal to the internal electromagnetic power for the subtransient stator equations used here. The constant mechanical torque is then set to this value at the operating point.

9.4 Sauer–Pai flux coupling

The transient and subtransient states are coupled through the Sauer–Pai γ factors. In generator-current convention the subtransient internal voltages are formed as

$$E_q'' = \gamma_{d1} E_q' + (1 - \gamma_{d1})\psi_d'', \qquad E_d'' = \gamma_{q1} E_d' + (1 - \gamma_{q1})\psi_q'',$$

with

$$\gamma_{d1} = \frac{X_d'' - X_l}{X_d' - X_l}, \qquad \gamma_{q1} = \frac{X_q'' - X_l}{X_q' - X_l}.$$

The derivative terms use the corresponding γ_{d2} and γ_{q2} coefficients on the same per-unit base as the state variables. This gives a consistent operating point for both the stator algebraic equations and the flux differential equations.

9.5 Per-unit base scaling

The network is on the 100 MVA system base, while the generator dynamic data are on the 900 MVA machine base. In this case the transformer per-unit data already account for the voltage-base change, so the machine reactances used inside the DAE are converted by the MVA ratio

$$X_{\text{system}} = X_{\text{machine}} \frac{100}{900}.$$

The γ_{d2} and γ_{q2} coefficients are then computed using the converted reactance differences so that the flux states and currents remain on one consistent base.

9.6 Current status

The 6th-order model now runs end-to-end:

- Power flow converges (6 iterations, min $|V| = 0.949$ pu).
- The DAE residual after refinement is below 5×10^{-14}.
- The local swing and damper modes are close to Kundur Table E12.3 (for example, modes near 1.06–1.12 Hz and real modes around -3 to -5 and -30 to -38).
- The remaining item for teacher review is the interarea damping: the computed interarea frequency is close to the textbook value, but the real part is still sensitive to the angle-reference and load/damping modelling assumptions.

10 Conclusion

The in-house Newton–Raphson power-flow solver converges for the Kundur Example 12.6 system in 6 iterations, yielding a minimum bus voltage of 0.949 pu and 85.1 MW of active losses. Under manual excitation the small-signal benchmark reproduces Kundur Table E12.3: the interarea mode at 0.545 Hz has a damping ratio of only 0.032, while the two local modes near 1.09–1.12 Hz have damping ratios of about 0.072. leaving a clean classical/manual-excitation comparison against the textbook.